

ROBOTICS (29/06/2021)

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The exam will be graded IFF the following recommendations have been taken into account:

- Use only the space provided, you can use scratch paper
- Write clearly so that the teacher can easily understand your answers
- Write your name, surname, and student id on scratch paper if you turn it in
- You are not allowed to use any programmable device (e.g., smartphone, calculator, etc.)
- You can use pen or pencil, paper will be provided, you cannot use notes or books

NAME: _____

SURNAME: _____

Student Id: _____

Exercise 1 – Sensors and Actuators [2+2 points]

Consider a classical LIDAR sensor used in mobile robots for localization and obstacle avoidance.

- 1.1: Describe the Beam Sensor Model for a LIDAR sensor and its parameters.

The beam sensor model uses a linear combination of 4 functions in order to evaluate the position z of an object that is expected to be found at $z = z_{exp}$. These functions are:

$f_1(z) = \begin{cases} \eta \lambda e^{-\lambda z} & \text{if } z \leq z_{exp} \\ 0 & \text{otherwise} \end{cases}$ this represents the fact that the LIDAR might detect an obstacle before the object

$f_2(z) = \eta \frac{1}{\sqrt{2\pi}b} \exp(-\frac{1}{2}(\frac{z - z_{exp}}{b})^2)$: represents a Gaussian around the point where we expect the object to be

$f_3(z) = \eta \frac{1}{z_{max}}$ where z_{max} is the maximum range of the LIDAR: represents the noise (2)

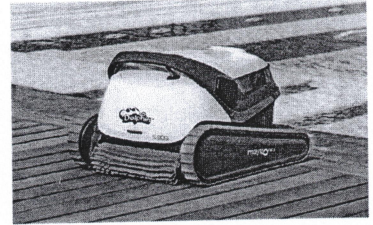
$f_4(z) = \eta \frac{1}{z_{small}}$ a function very high around z_{max} : it represents the probability to not detect the object but "detect" the maximum range instead; η is only for normalization.

- 1.2: Describe the Scan Sensor Model and its advantages w.r.t. the Beam Sensor Model

The Scan Sensor Model uses prior information about the map and the ^{pos} position in order to estimate the position of obstacles and objects detected by the sensors. The advantage is that it is less sensitive to sensor biases; the problem is that it introduces new sources of error (2)

Exercise 2 – Robot Odometry [2+1+3 point]

The Dolphin S200 is an electric robot for the cleaning of swimming pools. As you can see from the picture it has a peculiar kinematics. With reference to its kinematics answer the following questions:



- 2.1: What are direct and inverse kinematics and what are their uses?

The kinematics is the computation of the pose of the robot with respect to the parameters it depends on. It is used to know the future position of the robot when changing those parameters (here, the wheels of the tracks).

Inverse kinematics is the opposite approach: considering a given pose of the robot, we compute the right ~~parameters~~ values of the parameters in order to achieve it.

If

- 2.2: Provide the approximated direct kinematics equations for the Dolphin S200 swimming pool cleaner.

We have approximately: $V = \frac{V_1 + V_2}{2}$ where V_1 and V_2 are respectively the speeds of the right and the left track. We also have: $\omega = \frac{V_1 - V_2}{L}$ and $V_{xc} = V$ and $V_{yc} = 0$ because the robot is not capable of lateral movement. We have: $\alpha(t) = \int_0^t \omega(t') dt' + \alpha(0)$ and $\theta(t) = \int_0^t \omega(t') dt' + \theta(0)$.

- 2.3: Assume the robot has an equivalent wheelbase of 1.0 m and a maximum speed of 1.0 m/s compute the inverse kinematics to move from the initial position $[1, 1, 0]$ to $[4, 3, \pi/2]$, and then to $[4, -2, \frac{3}{4}\pi]$.

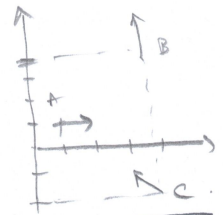
From A to B. Let the name of the points be: $A = [1, 1, 0]$, $B = [4, 3, \frac{\pi}{2}]$, $C = [4, -2, \frac{3}{4}\pi]$.

If the robot is in A at $t=0$, it should rotate until $\theta = \frac{\pi}{4}$.

$$\text{But } \theta(\tau) = \int_0^\tau \omega(t) dt + \theta(0), \text{ so } \frac{\pi}{4} = \frac{V_1 - V_2}{L} \tau = \frac{\pi}{4}.$$

$$\text{which means } \tau = \frac{\pi/4}{\frac{\pi}{4}} = 1.$$

So for $\tau = 1$ s, the robot should move anticlockwise: $V_1 = V_{max}, V_2 = -V_{max}$.



Then it goes in a straight line for τ_2 such that $\theta(\tau_1 + \tau_2) = \frac{3\pi}{4}$.

$$\text{and } \alpha(\tau_2) = \int_0^{\tau_2} \omega(t) dt + \alpha(\tau_1) \text{ where } \alpha(\tau_1) = \frac{\pi}{2}.$$

$$\text{So we have: } \tau_2 + \frac{\pi}{2} = \frac{3\pi}{4} \text{ then } \tau_2 = \frac{\pi}{4} \text{ s.}$$

The robot moves in a straight line to reach B for $\tau_2 = \frac{\pi}{4}$ s.

B → C Then it turns anti-clockwise by $\frac{\pi}{4}$ the time it takes was already computed: $\tau_3 = \tau_1 = 1$ s.

Then, the robot gets ready to go towards C, so it turns by π anticlockwise during $\tau_4 = \tau_1 = 1$ s.

Then, it moves in a straight line for $\tau_5 = 6$ s. $V_1 = V_2 = 1 \text{ m/s}$.

At last, it turns clockwise ($V_1 = -V_{max}, V_2 = V_{max}$) during $\tau_6 = 3\tau_1 = \frac{3\pi}{8}$ s.

Exercise 3 – Robot Local Path Planning [2+2+1]

Let's consider the Dynamic Window Approach for Local Path Planning.

- 3.1: Describe the DWA algorithm

The DWA is comprised of the following steps:

1. Sampling of the configuration space (in 2D) ?
2. Forward simulation for each speed sample for a ^{short} little time to see what happens on each trajectory.
3. Ranking all the trajectories by giving each of them a score.
4. Discarding the illegal ones (the ones that hit obstacles) and picking the one with the highest score.

DWA helps avoiding obstacles.

- 3.2: How does the DWA with trajectory roll-out differ from the standard DWA algorithm?

The DWA with trajectory roll-out differs from the standard DWA algorithm by the fact that it does not consider only movements in a 2-dimensional space, which makes up for a more accurate turning of the trajectory.

- 3.3: What is a navigation function and how it is used with DWA?

A navigation function is a function which allows to take into account heading, speed, goal, wheel direction... when planning a new trajectory. It is useful because we want to minimize the variations of these quantities when planning a trajectory in DWA.

Example of navigation function: $G = a \cdot \text{heading} + b \cdot \text{goal} + c \cdot \text{speed} + \text{wheel direction}$

Exercise 4 – Robot Operating System [3 points]

Answer the following questions:

- 4.1: What is ROS? ROS means "Robot Operating System". It is a software that provides several tools to program a robot, such as services, nodes, messages...

- 4.2: What are nodes? Nodes are small programs used in ROS ~~to program~~ in order to control the robot. They ~~comp~~ perform tasks such as computing odometry or ~~sending~~ sending movements to the wheels, etc. They communicate with one another by using messages, actions and services.
- 4.3: What are topics?

Topics are ~~exchange~~ ^{communication} channels on which messages are ~~trans~~ exchanged between nodes.

Topics have generally a publisher that publishes new content at a given rate and one or more subscribers that receive these messages as they are available. ROS offers several types of messages and new ones can be defined.

Exercise 5 – Localization and Mapping [1+2+2]

Let's consider the Simultaneous Localization and Mapping problem; answer the following questions:

- 5.1: Provide the derivation of the recursive Bayes Filter

The Bayes Filter is the following:

If the data received is an observation z :

For all x do:

$$\text{Bel}'(x) = P(z|x) \text{Bel}(x).$$

~~Normalize Bel'.~~

else: if it is an action a :

For all x do:

$$\text{Bel}'(a) = \int P(a|x',a') \text{Bel}(a') da'.$$

Return $\text{Bel}'(x)$.

?

- 5.2: How does the EKF SLAM algorithm implement the recursive Bayes Filter for online SLAM?

The EKF SLAM algorithm implements the recursive Bayes Filter by the fact that in order to provide a new estimate of the pose, it takes into account the prior belief with respect to the known (previous) pose as well as new observations.

0.5

- 5.3: When would you use the EKF SLAM algorithm? Provide a simple example and motivate it.

I would use the EKF SLAM in a situation where the map is not fully known but I can rely on a few landmarks to determine the position of the robot. One example could be a ~~work~~ construction site, where the map is not known and often changing. The robot would still have a few landmarks, like the corners, the existing walls, etc--

After exploring the environment, the robot, through the ^{Extended} Kalman Filter should have a good representation of its surroundings.

2

Exercise 6 – Planning [3]

Let's consider the A* algorithm used for Global Planning.

- 6.1: Apply the A* algorithm to the problem in the picture and provide the list of expanded nodes at each iteration (use 4 cells connectivity and the Manhattan distance as heuristics, e.g., from start to goal there are 9 steps)

00 Start	01	02	03	04	05	06	07	08
10	11	12	13	14	15	16	17	18 Goal
20	21	22	23	24	25	26	27	28
30	31	32	33	34	35	36	37	38
40	41	42	43	44	45	46	47	48

~~00~~

~~10~~

~~20~~

~~30~~

~~40~~ 20 22

10 20 22 32

10 20 32 13 24

10 20 32 13 14

10 20 32 13 15

10 20 32 13 14 05

10 20 32 13 14 05 06

10 20 32 13 14 05 07

10 20 32 13 14 05 07 08

10 20 32 13 14 05 08

(30)?

no h?

1.5

18

← the goal has been reached ; the path is:

00 → 01 → 11 → 21 → 22 → 23 → 24 → 14 → 15 → 05 → 06 → 07 → 17 → 18